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Yingliu

I - A

The World Still Needs Plenty of Turbine Blades: Read-Through to Yingliu

Following our preliminary thoughts on the impact of SpaceX's turbine blade plan ([report](#)), we conduct a deeper review of the turbine blade industry, leveraging on our US Aerospace & Defense analyst Seth Seifman's report: [Howmet Aerospace: The World Needs Plenty of Turbine Blades; Reit OW \(Sept 2, 2026\)](#). SpaceX's in-house casting plan is mainly a self-help response to power constraints on its AI compute buildout, rather than an attempt to supply third parties. We see limited risk to Yingliu, as casting capacity remains tight, qualification takes years and customer relationships are program-specific. We stay OW as Yingliu trades at ~32x/20x FY27E/28E P/E, and see its growth being supported by structural demand and continued customer expansion.

- **SpaceX is securing internal supply, not entering the broader market.** Its vertical integration and acquisition of APR Energy reflect concerns that power shortages could constrain its AI expansion. Commercializing high-temperature alloy blades at scale requires specialized equipment, stable production yields and multi-year development and qualification. Even if SpaceX successfully meets part of its own demand, incumbent suppliers will still need to serve utilities, data centers and industrial customers. The initiative therefore reinforces the view that power needs and turbine-blade demand will remain elevated.
- **Tight supply and high barriers support a long upcycle.** Doncasters forecasts global gas turbine airfoil casting orders to grow at a 9% CAGR during 2024-35, with heavy-frame airfoils growing faster. New capacity is slow to add, given 12-18-month equipment lead times, 40-50 tightly controlled production stages and multi-year OEM qualification. Production yields can range from 50%-90%, further limiting the industry's ability to respond quickly to rising demand and supporting sustained supply tightness.
- **Yingliu's customer ramp is gaining visibility.** Siemens Energy development now covers the 4000F, 8000H and 9000HL platforms, while Baker Hughes renewed its agreement for 5 years and expanded Yingliu's role from casting into machining and coating. Ansaldo orders ramp, Doosan's 380MW blade program continues to progress, and Safran and Rolls-Royce qualifications support a second growth leg in aero engines.

SpaceX's turbine blade plan

This section refers directly to the analysis of our US Aerospace & Defense analyst Seth Seifman. See Seth's full report: [Howmet Aerospace: The World Needs Plenty of Turbine Blades; Reit OW \(Sept 2, 2026\)](#).

SpaceX's move is mainly self-help, not disruption. Drawing directly from Seth's report, SpaceX's in-house blade casting is a self-help measure to address its own power and AI compute constraints, not a disruptive threat to the established supply chain. SpaceX's vertical integration reflects its belief that power and chip shortages could slow its AI buildout. The APR Energy acquisition and blade

Overweight

603308.SS, 603308.CH

Price (03 Sep 26): Rmb43.18

Price Target (Dec-27): Rmb85.00

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casting plans are about securing internal supply, not competing with blade manufacturers like Howmet.

Seth's report emphasizes the gas turbine industry is already booked out for years, and SpaceX's efforts are unlikely to change the durable demand for Howmet, which remains the leading blade manufacturer. Howmet's capex and supply agreements with seven major industrial gas turbine customers underpin its growth, with some customers' backlogs into the 2030s. Even if SpaceX produces blades for itself, the market will remain supply-constrained, and Howmet's products are not commoditized—high prices will not drive rapid market saturation.

In response, Yingliu's management said that they expect bottlenecks to persist for 3–5 years, with high-temperature alloy casting capital-intensive and slow to commercialize. Yingliu views SpaceX's initiative as focused on internal needs, not a threat to global supply or its own overseas orders.

Turbine blade 101

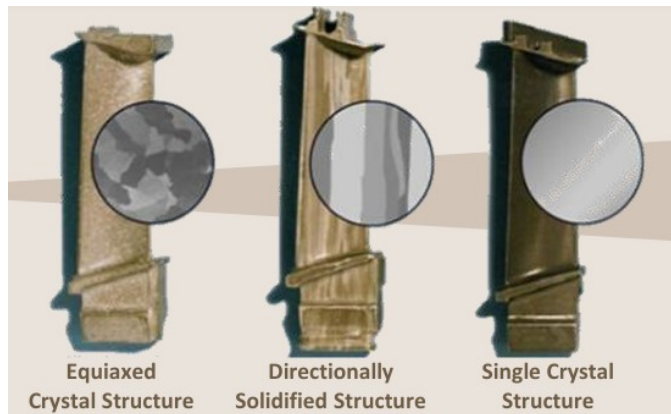
What is a blade? A turbine blade is a precision-made airfoil that turns hot, high-pressure gas into rotational power. In an industrial gas turbine, this power drives a generator. In a jet engine, it turns the shaft and fan to help produce thrust. Blades are critical because engine efficiency, durability and economics depend heavily on how well the turbine section handles higher temperatures.

Types of blades by structure: **1) Single-crystal blades:** Made from one crystal with no internal grain boundaries, they provide the strongest resistance to heat, stress and fatigue. They are used in the hottest sections of advanced jet engines and gas turbines; **2) Directionally solidified blades:** Their grains run in aligned columns, improving strength and blade life versus conventional castings. They are widely used in demanding turbine stages; **3) Equiaxed blades:** Their grains face different directions, making them less heat-resistant and better suited to less demanding sections.

Why are blades so demanding? **1) For jet engines**, at aircraft take-off, turbine temperatures can >1,600°C, around 450°C above the melting point of the underlying alloys. Blades may rotate at 12,000-15,000 rpm and face pressure ratios of ~40:1. They survive through nickel-based superalloys, internal cooling passages, film-cooling holes and heat-resistant coatings. **2) For turbine blades**, they use many of the same alloys, cooling designs and casting methods as aero blades, but can be measured in feet rather than inches. Manufacturers must therefore maintain similar precision across much larger parts.

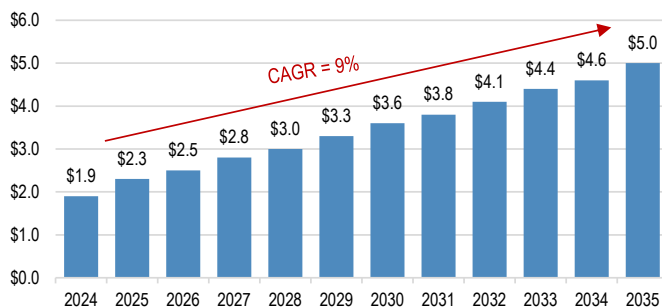
How fast is the market growing? Doncasters forecasts global industrial gas turbine airfoil casting orders to grow 9% annually from 2024 to 2035, with heavy-frame airfoils growing faster. Major industrial gas turbine OEMs have a combined backlog of over 170GW, equal to 3-4 years of production visibility. In aerospace, Howmet cites >10 years of aircraft backlogs. Our separate estimate based on channel checks suggest that the value of the global gas-turbine market is over US\$20B annually, with blades representing ~20%, or c.US\$4B.

Figure 1: Blade types by structure



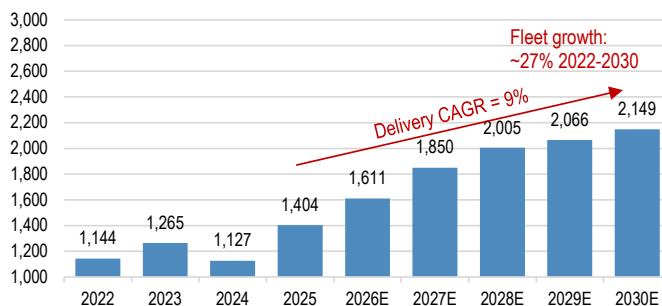
Source: Howmet presentation.

Figure 2: Industrial gas turbine airfoil castings market size and growth (US\$)



Source: IEA and Doncasters. IGT airfoil castings generally include blades and vanes.

Figure 3: Commercial aero deliveries expected to grow double digits through 2027



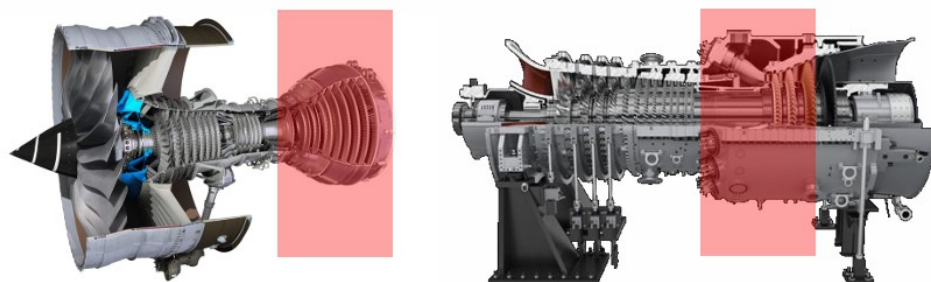
Source: Cirium data. Howmet presentation.

Why are the blades so difficult to manufacture? Turbine blades operate at ~1,000-2,000°C and require advanced alloys and specialized single-crystal or directionally solidified casting. Production involves 40-50 stages, yields can be ~50-90%, and OEM qualification takes years. New equipment also needs 12-18 months before installation and ramp-up, keeping global supply concentrated and tight.

Industry landscape: The market is dominated by a handful of global players—Howmet, PCC (Precision Castparts Corp), Doncasters, CPP (Consolidated Precision Products) and a few emerging suppliers like Yingliu—due to the formidable technical, capital, and qualification barriers. Yingliu currently holds only ~3% of the global gas turbine blade market.

Figure 4: What Yingliu is producing

Jet engines (left hand side); gas turbines (right hand side). Red areas: superalloy hot-section parts for aero engines and gas turbines, where Yingliu is producing.



Source: Yingliu's presentation.

Figure 5: Blade and vanes



Source: Yingliu company website.

Figure 6: Heavy-Duty gas turbine blade



Source: Howmet website.

Table 1: Global gas turbine components and OEM peer comparison

Components	BBG Ticker	JPM Rating	Last Price LC	JPM PT LC	Upside/ Downside	Mkt Cap US\$m	Daily turnover US\$m	YTD Stock perf.	P/E (x)			EPS growth Y/Y			PEG	
									FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E
Yingliu	603308 CH	OW	43	85	97%	4,681	173	3%	50	32	20	69%	59%	57%	0.8	0.6
Howmet	HWI US	OW	253	350	38%	100,881	738	23%	47	39	33	46%	20%	19%	2.3	2.0
Doncasters	DPC US	NC	39	NA	NA	5,897	75	NA	66	43	35	NA	54%	24%	1.2	1.8
Wedge	000534 CH	NC	24	NA	NA	1,834	76	10%	39	27	19	31%	44%	42%	0.9	0.6
Himile	002395 CH	NC	42	NA	NA	7,299	40	-27%	17	15	13	13%	16%	19%	1.1	0.8
Average									44	31	24	40%	39%	32%	1.3	1.2
OEMs																
GE Vernova	GEV US	OW	922	1,330	44%	245,544	2,738	41%	39	38	27	220%	3%	41%	14.5	0.9
Siemens En	ENR GY	OW	144	245	70%	144,361	397	20%	27	20	16	225%	35%	23%	0.8	0.9
MHI	7011 JP	NC	3,795	NA	NA	81,756	466	-1%	45	30	25	6%	51%	17%	0.9	1.7
Baker Hughes	BKR US	OW	65	74	14%	64,157	484	42%	25	23	18	4%	12%	27%	2.1	0.8
Doosan Enerbility	034020 KS	OW	79,500	130,000	64%	37,530	324	5%	127	76	51	N.A	67%	47%	1.9	1.6
Caterpillar	CAT US	OW	792	1,100	39%	364,191	2,836	38%	28	24	20	51%	18%	20%	1.5	1.2
Jerih	002353 CH	OW	124	162	31%	18,886	431	75%	37	25	23	14%	47%	7%	0.8	3.5
Dongfang Elec	1072 HK	NC	20	NA	NA	12,189	28	-21%	12	11	9	20%	15%	20%	0.8	0.5
Shanghai Elec	2727 HK	NC	3	NA	NA	13,378	10	-23%	28	21	15	86%	33%	37%	0.8	0.6
Average									41	30	23	78%	31%	27%	2.7	1.3

Source: Bloomberg Finance L.P., J.P. Morgan estimates. Estimates for Not Covered (NC) companies are based on Bloomberg consensus data. Japanese names have financial years ending March. DPC was listed only in June 2026. Priced at close of September 3, 2026.

Related reports

- [Yingliu: Thoughts on SpaceX's turbine blade plan and its impact \(September 1, 2026\).](#)
- [Yingliu: 1H26 Briefing Takeaways: Demand intact, guidance maintained; Model update \(August 31, 2026\).](#)
- [Yingliu: 2Q26 review: Cost pressure weighs; demand and backlog momentum intact \(August 30, 2026\).](#)
- [Yingliu: Read-across from Siemens Energy 3Q'26 results \(August 5, 2026\).](#)
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- [Yingliu: Factory visit takeaways \(July 13, 2026\).](#)
- [Yingliu: JPM Fireside Chat Takeaways \(June 4, 2026\)](#)
- [Yingliu: Rare China play on the global gas turbine and aero-engine bottleneck; initiate with OW \(June 1, 2026\).](#)

Investment Thesis, Valuation and Risks

Yingliu

I - A (Overweight; Price Target: Rmb85.00)

Investment Thesis

Yingliu is China's largest high-temperature alloy and precision casting supplier positioned across gas turbine and aero-engine hot-end components, with access to global OEMs including Baker Hughes, Siemens Energy, Ansaldo, Doosan, GE Aerospace, Safran and Rolls-Royce. Its core appeal is scarcity: Yingliu's FY25A gas turbine revenue was only Rmb1B, implying about 2% global blade market share, but we expect this to rise to 10% by 2030 as OEMs diversify suppliers. Gas turbines are the near-term growth engine, supported by AIDC power demand and industry undersupply, with global gas turbine demand expected to exceed 100GW annually from 2026 to 2035, while 2026 manufacturing capacity is only 64GW. Aero engines provide a second growth leg, as Yingliu is the only China-based hot-end supplier to GE Aerospace, Safran and Rolls-Royce, with long-term contracts extending to 2030–2032.

Valuation

We derive our Rmb85 Dec-27 PT by applying 40x P/E to Yingliu's FY28E EPS, using Howmet's FY27E P/E of around 40x as the key benchmark and noting our US analyst's 45x FY28E target multiple for Howmet.

Risks to Rating and Price Target

Downside risks include: 1) slower-than-expected gas turbine demand growth if alternative power solutions such as gas engines gain share; 2) delayed customer ramp-up at key OEMs, including Baker Hughes, Siemens Energy and Ansaldo; 3) lower-than-expected GPM improvement due to weaker production yield, pricing or product mix; and 4) geopolitical or export-control risks affecting Russian client exposure or access to imported US/EU equipment needed for capacity expansion.

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Yingliu Electromechanical - A (603308.SS, 603308 CH) Price Chart



Date	Rating	Price (Rmb)	Price Target (Rmb)
01-Jun-26	OW	64.67	95
31-Aug-26	OW	43.99	85

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jun 01, 2026. All share prices are as of market close on the previous business day.

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